

ADITYA SHIRSATRAO

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Solapur, Maharashtra, India

PROFESSIONAL SUMMARY

B.Tech AI & Data Science student (CGPA 8.5/10) with hands-on experience building production-grade ML systems, NLP pipelines, and full-stack AI products across cybersecurity, observability, and legal-tech domains. Three-time national hackathon winner, IEEE first-author publication at ICCTWC 2026, and GDG AI/ML Lead mentoring 200+ students. Proficient in PyTorch, TensorFlow, HuggingFace, and RAG/LLM systems.

EDUCATION

N. K. Orchid College of Engineering & Technology

B.Tech in Artificial Intelligence and Data Science – **CGPA: 8.5/10**

Solapur, Maharashtra

Aug. 2023 – Present (Expected 2027)

TECHNICAL SKILLS

Machine Learning: PyTorch, TensorFlow 2.x, HuggingFace Transformers, Scikit-learn, OpenCV, NLP, Computer Vision, Transfer Learning, Model Optimization (Quantization, Distillation), RAG, LangChain, AI Agents, FAISS, Vector Embeddings

Programming Languages: Python (expert), Java, C++, JavaScript, SQL

Backend & Infrastructure: FastAPI, Node.js, PostgreSQL, Redis, Celery, Docker, Kubernetes, Kafka, Microservices

Core Competencies: DSA (366+ LeetCode), System Design, ML Systems Architecture, Anomaly Detection, Feature Engineering, Model Evaluation, Real-time Data Processing

EXPERIENCE

Fullstack Web Development Intern

Unified Mentor Pvt. Ltd

Production ML & Distributed Systems Development

Jul. 2025 – Jan. 2026

- Architected an **AI-powered code review platform** with asynchronous GitHub PR processing; integrated Kafka queues and Redis caching, reducing analysis latency by **45%** via parallel distributed workers and vectorized AST-based code search.
- Built a **distributed ML inference platform** with autoscaling endpoints, model versioning, and Prometheus monitoring; improved throughput by **60%** under concurrent workloads via request batching and inference caching.

AI/ML Lead

Google Developer Groups on Campus (GDG-NKOCET)

Community Leadership & Mentorship

Sep. 2025 – Present

- Lead AI/ML initiatives for a **200+ student** community, organizing workshops on production ML, LLMs, and MLOps best practices.

ML ENGINEERING PROJECTS

SentinelX – AI Cybersecurity Platform | *Python, FastAPI, DistilBERT, Whisper, Celery, Redis, PostgreSQL* | Code May 2026

- Built a production-grade multi-channel threat detection platform analyzing phishing, scams, and social engineering across email, SMS, and voice using a multi-model pipeline: HuggingFace DistilBERT for NLP classification, OpenAI Whisper for speech-to-text, and Qwen 2.5-7B for LLM enrichment.

– Engineered a weighted 4-component risk scoring engine (NLP, behavioral, URL, and sender reputation signals) into a unified threat score, decoupled via async Celery + Redis task queues; won **1st Place at Hack to Future 3.0** (600+ teams, INR 50,000).

RAG Knowledge Assistant | *FastAPI, Chroma, sentence-transformers, BM25, LangChain* | Code Jun. 2026

- Built an LLM Q&A system over 500+ documents using hybrid retrieval (dense vector + BM25) with source citations, achieving **92% retrieval hit-rate** on a 50-question eval set; runs offline with zero model downloads.

Production ML System: Network Intrusion Detection | *TensorFlow, scikit-learn, ML Systems Design* | Code Mar. 2026 – Apr. 2026

- Architected a complete ML system for real-time anomaly detection and multi-class threat classification on streaming network traffic.
- Engineered a feature pipeline extracting **40+ features**, achieving **96% accuracy** with **<100ms** inference latency on NSL-KDD and UNSW-NB15 (24+ attack types), **95%+ recall** on novel attacks.

– Handled class imbalance with SMOTE; compared Random Forest, XGBoost, and LSTM for the precision-recall trade-off.

Multilingual Sentiment Analysis with BERT Fine-Tuning | *PyTorch, HuggingFace Transformers, NLP* | Code Apr. 2026

- Fine-tuned BEKT-base-multilingual-uncased for 104+ languages achieving **92%+ accuracy** on multilingual sentiment classification; published the model to the HuggingFace Model Hub.
- Applied 8-bit quantization, knowledge distillation, and batch processing, reducing inference latency by **35%** while maintaining accuracy.

ACHIEVEMENTS & RESEARCH

IEEE First-Author Publication on multilingual sentiment analysis and cross-lingual classification, presented at ICCTWC 2026.

Hackathon Champion: 1st Place Hack to Future 3.0 (600+ teams); 1st Runner-Up Orchathon 2K26 (800+ teams).

Open Source: 50+ PRs merged across Google (OSS-Fuzz), Microsoft, Apache (DataFusion), and Hedera (Hiero SDK).

Certifications: Kaggle AI Agents; JPMorgan Chase & Electronic Arts SWE Virtual Experience; NPTEL Java (Elite+Gold, 91/100).